

Design and Technology Co-optimization Utilizing Flip-FET (FFET) Standard Cells

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Abstract—With the continued scaling of VLSI technology beyond 3nm, a consistent demand for layout reduction in standard cells has been made. CFET (Complementary-FET) has been accepted as a promising device technology, stacking N-FET on P-FET (or vice versa) to achieve this goal while providing metal interconnects on both the front and backside of the wafer through BEOL (back-end-of-line) processing. However, the layout synthesis of CFET based standard cells and its use in physical design implementation are not fully compatible with the effective exploitation of backside interconnects. This is because of a considerable overhead on the allocation of special vias in FEOL (front-end-of-line) referred to as *tap-cells*, which are essential for the net routes using backside wire. To overcome this drawback of using CFET cells, a new technology called FFET (Flip-FET) has been proposed, which *flips* the lower FET in CFET to enable direct pin accessibility on both sides of BEOL with no tap cells. In this context, we propose an FFET cell based DTCO methodology to fully utilize backside wires with minimal tap-cells. Precisely, we propose a three-step approach: (1) synthesizing multiple FFET standard cells with diverse styles of pin distribution for each of primitive logic gates, (2) performing a tap-cell avoiding cell replacement during placement optimization by using the cells obtained from Step 1 to prevent tap-cell allocation for net routing, and (3) rebalancing the wire usage between the frontside and backside to mitigate net congestion. Experiments with benchmark circuits show that our FFET cell based DTCO methodology scales up chip size beyond CFET designs with much fewer routing failures.

Keywords: CFET, FFET, standard cells, DTCO, physical design

I. INTRODUCTION

With aggressive advancement in technology nodes which drives FinFET to reach its structural limits in scaling, interest has shifted toward vertically stacked 3D architectures for further scaling. One noticeable technology innovation is Complementary FET (CFET) [1], which physically stacks N-FET on top of P-FET or vice versa (e.g., Fig. 1(b)), for which installing backside power delivery network (BSPDN) with buried power rails (BPR) [2] is essential to secure a few more metal tracks available at the frontside for net routing.

Despite the use of BSPDN, chip routing for designs with CFET based standard cells still suffers a considerable shortage of metal resources. To tackle this issue, a recent work [3] and TSMC patent [4] suggested using backside wires not only for BSPDN but also for signal net routing.

However, for chip-level double-sided routing, whenever a net uses a backside wire to make a route, it is unavoidable to allocate a special via called *tap-cell* to connect the net pins to the backside wire since all cell pins are exclusively located on a metal layer at the frontside. Thus, if a balanced

use of double-sided metals were made for the routes of n nets, it is required to allocate roughly $n/2$ tap-cells, which is an enormous overhead on chip PPA. (Note that the works in [5], [6] proposed a netlist planning that distributes nets across both sides to mitigate the occurrences of routing failure. On the other hand, the work in [7] utilized backside wires for ECO routing. However, all the works in [5]–[7] inevitably suffered from the tap-cell overhead.)

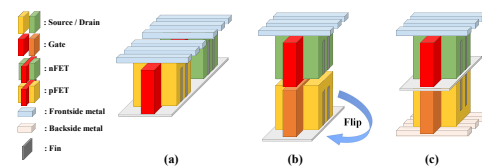


Fig. 1. Structures of FinFET, CFET, and FFET. (a) FinFET: routing tracks only at the frontside. (b) CFET: much smaller footprint than (a), but routing tracks only at the frontside. (c) FFET: a little smaller footprint than (b) and routing tracks at both the front and back sides.

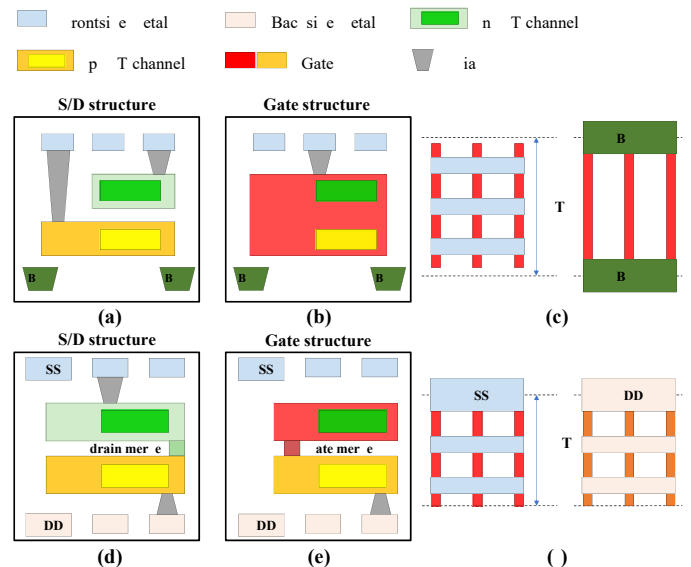


Fig. 2. Cross sectional view of 3T-height CFET ((a), (b), (c)) and 2.5T-height FFET ((d), (e), (f)).

Recently, a new 3D transistor technology called Flip-FET (FFET) [8] has been proposed, which *flips* the lower FET in CFET such that the locations of cell I/O pins can provide direct pin accessibility through either frontside or backside BEOL with no tap cells. Fig. 1(c) shows an illustrative FFET structure, in which two layers of FETs are formed and stacked

back-to-back, offering a manufacturing-compatible process flow with a lower aspect ratio in comparison to that of CFET structure in Fig. 1(b). Fig. 2 shows the structural differences between FFET and CFET with three horizontal metal tracks, in which each of the front and back sides in FFET runs a single power rail and 2 RTs (routing tracks), resulting in a total of 4 RTs and 2.5T cell height¹ while CFET runs 3 RTs only at the frontside, resulting in 3T cell height.

Note that regarding the automatic generation of standard cell layouts, many algorithms and frameworks have been proposed in the literature. Some of the representative works on FinFET based cell synthesis are found in [9]–[12], and the DTCO (design technology co-optimization) process using a cell generator as a core engine is well described in [13]. In addition, CFET based cell synthesis methods with SMT (satisfiability modulo theory) formulation and with an exploration of dynamic transistor foldings are proposed in [14], [15], and [16], respectively. However, none of the prior works have taken into account the exploitation of backside wires to improve pin accessibility, thus taking no consideration of tap-cell overhead at all.

In this work, we propose an FFET cell based DTCO methodology to fully make use of backside wires with minimal use of tap-cells. The contributions of our work can be summarized as:

1. We propose an *effective method of synthesizing FFET based standard cells*, the problem of which has never been addressed as yet, though a couple of automatic synthesis methods of CFET based standard cells have been proposed in the literature. Our cell generator exactly considers two differences in in-cell routing: (1) The flexible (front/back) side selection for individual I/O pin locations, and (2) The availability of additional metals i.e., the metals on BM1 (backside metal 1) layer.
2. We propose a *DTCO driven tap-cell avoiding cell replacement algorithm* that utilizes multiple cells with diverse styles of I/O pin distributions to eliminate the need for tap-cells during the placement optimization stage.
3. We propose a *net redistribution method*, which determines metal sides for all nets whose routes can be made by using either the frontside metals exclusively or the backside metals exclusively, thus unnecessary to allocate tap-cells so that the net connection on both sides should be mitigated in an unbiased manner.
4. Through experiments with benchmark circuits, it is shown that our proposed FFET cell based DTCO methodology is very promising, scaling up the chip size further over that of the CFET cell based design implementations while incurring much fewer routing failures.

II. PRELIMINARY, MOTIVATION, AND NET CLASSIFICATION

CFET cell vs. FFET cell: We itemize the architectural differences between CFET [1] based and FFET [8] based cells

¹In FFET, one of the three horizontal metals is used as a power rail, which will be shared by neighboring cells.

in the following.

1. *Cell height:* For the same number of frontside metal tracks available for routing (e.g., 3 RTs (routing tracks) in Fig. 2), FFET allocates one track on each side for the power rails, allowing neighboring cells to share the power rails effectively. In contrast, CFET utilizes two backside tracks designated for BPR. Therefore, as illustrated in Fig. 2, CFET employs only three frontside RTs for BEOL processes, while FFET utilizes two RTs on each side (total of 4 RTs). As a result, FFET architecture has 0.5T smaller cell height while using the same amount of routing tracks in comparison with that of CFET architecture.
2. *Cell via:* CFET requires long (deep) vias to connect frontside metals to the FET in the lower layer, as illustrated in Fig. 2(a), introducing complex spacing rules. On the other hand, since FFET is formed by merely flipping the lower FET in CFET, the frontside FET is allowed to connect directly to the frontside metals and the backside FET to connect directly to backside metals.² This enables FFET to position I/O pins on either side. This flexible I/O pin positioning is particularly useful for making double-sided routing without tap-cells.
3. *Gate merge:* The implementation of split-gate structures allows for a reduction in the cell width, as shown in Figs. 3(b) and (c). FFET can make split-gate structures easily by *gate merge* with no area and wire overhead, as illustrated in Fig. 2(e). While split-gate CFET can also accommodate different signals on each gate, it requires inevitable area overhead. (Note that the split-gate of the lower FET in CFET is only accessible through frontside metals, resulting in additional spacing whereas that of FFET is accessible through backside metals.)

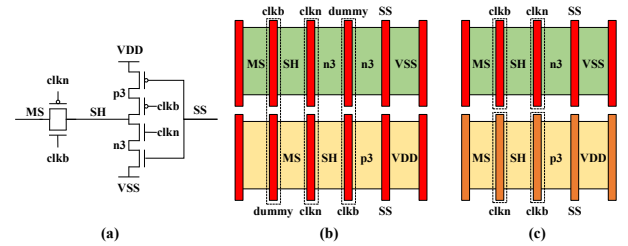


Fig. 3. Difference in gate split for CFET and FFET. (a) Netlist example: C^2MOS in DFFHQNx1. (b) Layout w/o split-gate on CFET: requiring dummy gate polys. (c) Layout with split-gate on FFET.

4. *Drain merge:* Both CFET and FFET can make *drain merge* with no overhead, as shown in Fig. 2. However, due to the fact that the output signal of the cell is always derived from a merged (common) drain, the signal can be delivered directly through metals on both sides for FFET, and it becomes possible to locate the output pin pattern on both sides to boost the pin accessibility as well as to minimize the number of tap-cell allocations.

²In this work, we assume the NFET on the frontside and the PFET on the backside, though they could be placed on either side.

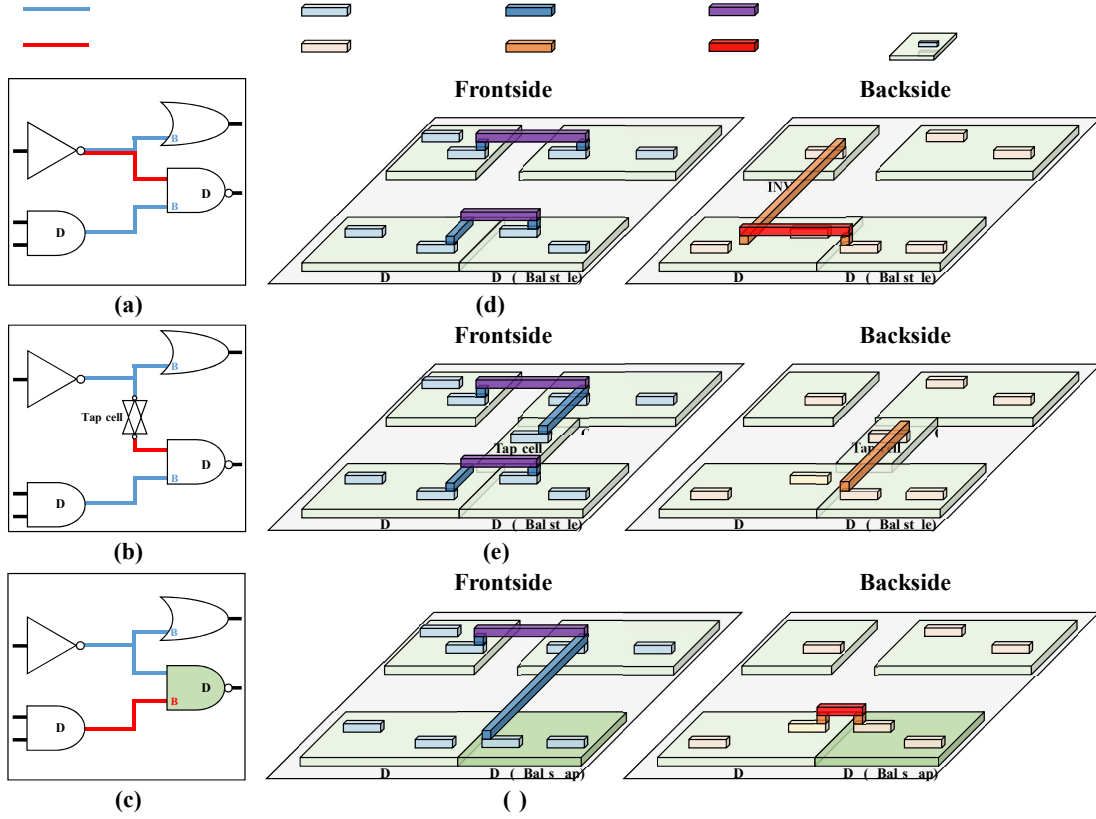


Fig. 4. Impact of BEOL side assignment for FFET cells' (input) pins on the number of tap-cells and WL (wirelength). Using NAND2 cell instances (a) with A on backside and B on frontside, (b) with A on backside and B on frontside connecting to a tap-cell (c) with A on frontside and B on backside. Feasible placement/routing results (d) WL = 37, #tap-cells = 0 for (a), (e) WL = 27, #tap-cells = 1 for (b), (f) WL = 21, #tap-cells = 0 for (c).

Motivation (tap-cells and WL): We describe, using an example, how the BEOL side (i.e., front/back sides) assignment of FFET cells' I/O pins affects the number of tap-cells and WL (wirelength) to be used. Figs. 4(a, d) and (b, e) show two different BEOL side assignments of input pins A and B on the NAND2 FFET cell in a part of the logic netlist where the OR2 cell commonly locates pin B on the frontside. Then, since output pins can be allocated to both sides, as observed in *item 4* of CFET vs. FFET in this section, the pin distribution of the NAND2 cell in Fig. 4(a, d) causes double-sided routing for net_1 , which otherwise, requires one tap-cell as shown in Fig. 4(b, d). On the other hand, the pin distribution of the NAND2 cell in Fig. 4(c, f) makes only frontside routing for net_1 with no tap-cell. Thus, the clever selection (i.e., side assignment for pins) of the NAND2 cell instance can reduce the resulting WL (from 37 to 21 in Figs. 4(a, d) and (c, f) and the number of tap-cells.

Net classification: As observed in Fig. 4, all the nets to route can be classified into three types:

- *Frontside-only net (FS-net):* A net whose all driven pins are accessed through the metals on the frontside (e.g., net_2 in Fig. 4(a) and net_1 in Fig. 4(c)).
- *Backside-only nets (BS-net):* A net whose all driven pins are accessed through the metals on the backside (e.g., net_2 in Fig. 4(c)).
- *Dual-side nets (DS-net):* A net that belongs to neither *FS-net* nor *BS-net* (e.g., net_1 in Fig. 4(a)).

Note that unlike the double-sided routing for designs with non-FFET cells (e.g., [4]–[7]), which assumes low resistance on backside wires to account for routing timing-critical nets only with backside wires, but no consideration of tap-cell delay, we assume exactly identical metal configurations on both sides. This symmetry offers a significant technological advantage, as it enables the reuse of the same process for both N-FET and P-FET, as described in [8].

III. THE PROPOSED DTCO METHOD

A. Overall DTCO Flow

Fig. 5 shows the overall flow of our DTCO framework, which is composed of three steps: **(Step 1)** we build up an FFET based standard cell synthesizer, which systematically generates multiple FFET cells of diverse styles of pin distribution for each primitive logic gate by considering the FFET's distinct feature that individual I/O pins on an FFET cell can be distributed on either side; **(Step 2)** at the placement optimization, we perform a tap-cell avoiding cell replacement by using the cells of diverse pin distributions prepared in Step 1 while minimizing metal resources and eliminating the need for allocating tap-cells simultaneously; **(Step 3)** following the cell replacement at the placement optimization, we perform a congestion-aware net rebalancing, so that the net congestion on both sides should be equally mitigated. After completing all steps, *FS-nets* as well as frontside segments of *DS-nets*

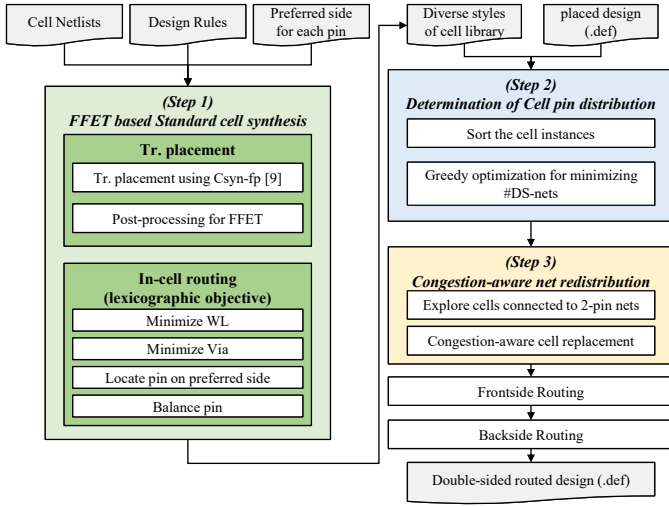


Fig. 5. Overall flow of our DTCO framework.

are routed at the frontside while *BS-nets* as well as backside segments of *DS-nets* are routed at the backside.

B. Synthesis of FFET based Standard Cells

1) **Transistor placement:** Given the cell's netlist information and design rules (e.g., OD jog rule) as inputs, as our initial transistor placements, we use the transistor placements produced by Csyn-fp [9] installed OpenROAD framework [17], which ensures cell area optimality. Then, we update the transistor placements by considering the following two factors: (1) locating cell output pins on both sides to support double-sided routing and (2) preferring *gate merge* to allow vertically merging dummy and non-dummy poly gates to increase the number of gate contact points.

2) **In-cell routing:** Our in-cell router generates a legal cell layout by incorporating design rules and the preferred BEOL side of individual input pins. As the conventional SMT formulation of in-cell routing, our SMT formulation includes, as constraints, representative design rules including minimum area rule (MAR), end-of-line (EOL) spacing, via enclosure (VE), and multiple patterning design rules such as parallel run length (PRL) and step height rule (SHR). In addition, our formulation uses the metals on *both backside and frontside metal layers*.

Our in-cell router is designed to optimize multi-objectives in a lexicographical order. The primary objective is to minimize metal length, followed by reducing the number of vias. The third objective prioritizes the input pin's preferred side, maximizing metal usage on that side. (If nothing is given as input, this objective is ignored.) The last objective is to balance the number of pins placed on both sides as double-sided routing could be the most effective when nets are distributed equally on both sides.

3) **Generate multiple FFET cells of diverse styles:** For each logic gate type, our cell synthesizer produces four styles of cell layout as follows.

- **Pin balanced (PBal):** The cell layout with balanced pin distribution on both sides (e.g., Fig. 6(a)).

- **Pin balanced-swapped (PBal-swap):** The cell layout with pin distribution sides on PBal swapped (e.g., Fig. 6(b)).
- **Pin-frontside (PFront):** The cell layout with all pins preferred on frontside (e.g., Fig. 6(c)).
- **Pin-backside (PBack):** The cell layout with all pins preferred on backside (e.g., Fig. 6(d)).

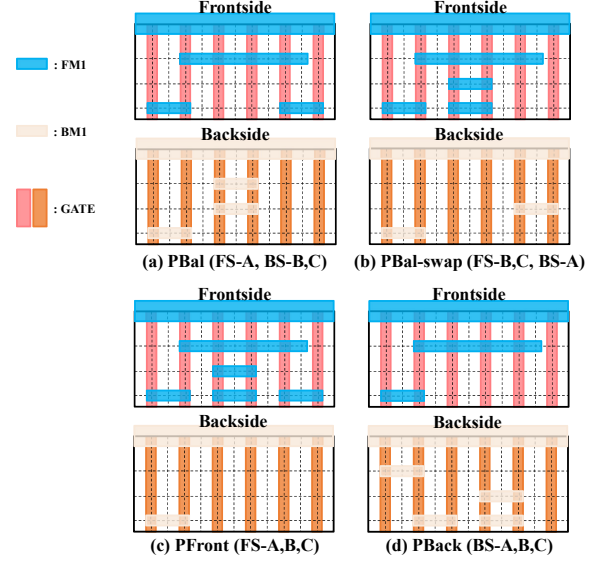


Fig. 6. Four styles of cell layout of OR3x1 generated by our cell synthesizer.

C. Determination of Cell Pin Distribution for WL Minimization

At the post-place, we determine, for each cell instance, the cell layout styles among **PBal**, **PBal-swap**, **PFront**, and **PBack** in a way to produce *DS-nets* (dual-side nets) as minimal as possible since routing *DS-nets* should use wires on *both* sides, incurring a longer total length of wires than that of *FS-nets* (frontside-only nets) and *BS-nets* (backside-only nets). To this end, we employ a greedy approach:

1. Sort the cell instances in placement in a non-increasing order according to the number of input pins. We break ties by the total amount of HPWL of the nets connecting the pins, placing priority on the cell instance with higher HPWL value.
2. For each cell instance c_i in the sorted list with input pins $p_1, \dots$, we compute cost $\#DS(c_i, c'_i)$ for each cell style $c'_i \in \{\text{PBal}, \text{PBal-swap}, \text{PFront}, \text{PBack}\}$:

$$\#DS(c_i, c'_i) = \sum_{j=1, \dots} \rho(c_i, c'_i, p_j), \quad (1)$$

where $\rho(c_i, c'_i, p_j) = 1$ if replacing c_i with c'_i enables the net connecting p_j to become *DS-net*, and 0, otherwise. Then, we select the cell style which has the smallest $\#DS(c_i, c'_i)$ value and perform cell replacement. For the same $\#DS(c_i, c'_i)$ value, we choose the cell style with the smallest HPWL value.

TABLE I

COMPARISON OF CELL METRICS BETWEEN CFET BASED 4T STANDARD CELLS PRODUCED BY CFET CELL SYNTHESIZER [16] AND OUR FFET BASED 3.5T STANDARD CELLS. #CPP, WL, M1, M2, M3 INDICATE THE NUMBER OF CONTACTED POLY PITCHES (I.E., CELL WIDTH), TOTAL WIRELENGTH, THE METAL USAGE ON M1, M2, M3, RESPECTIVELY.

Cell	#Cell	#track	CFET based 4T Standard Cells [16]							FFET based 3.5T Standard Cells (Ours)						
			Height	#CPP	Area	WL	M1	M2	M3	Height	#CPP	Area	WL	M1	M2	M3
AND	3	4	180	6.33	1140.00	71.00	16.00	11.00	0.00	141	6.33	893.00	58.67	24.00	0.00	0.00
AOI	2	4	180	10.00	1800.00	170.00	56.50	13.50	0.00	141	9.50	1339.50	126.00	50.00	0.00	0.00
BUF	4	4	180	7.50	1350.00	71.75	19.75	6.00	0.00	141	7.50	1057.50	70.50	26.50	0.00	0.00
DEFFHQN	3	4	180	16.00	2880.00	230.00	84.67	13.00	7.00	141	16.67	2350.00	231.00	111.00	0.00	0.00
DHL	3	4	180	12.00	2160.00	143.67	57.33	9.00	0.00	141	13.00	1833.00	161.33	70.67	0.00	0.00
INV	4	4	180	5.75	1035.00	50.25	13.25	6.00	0.00	141	5.75	810.75	50.00	19.00	0.00	0.00
NAND	4	4	180	12.00	2160.00	173.25	63.75	10.50	0.00	141	12.00	1692.00	152.25	64.25	0.00	0.00
NOR	4	4	180	12.00	2160.00	186.00	64.00	13.75	1.25	141	11.75	1656.75	149.50	65.50	0.00	0.00
OAI	2	4	180	10.00	1800.00	166.00	54.50	13.50	0.00	141	10.00	1410.00	131.00	51.00	0.00	0.00
OR	3	4	180	6.33	1140.00	71.00	16.00	11.00	0.00	141	6.33	893.00	58.67	24.00	0.00	0.00
XNOR	1	4	180	11.00	1980.00	204.00	63.00	16.00	9.00	141	12.00	1692.00	158.00	74.00	0.00	0.00
XOR	1	4	180	11.00	1980.00	188.00	58.00	13.00	9.00	141	12.00	1692.00	162.00	78.00	0.00	0.00
Avg.		4	180	9.99	1798.75	143.74	47.23	11.35	2.19	141	10.24	1443.29	125.74	54.83	0.00	0.00
Norm.		4	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.78	1.02	0.80	0.87	1.16	0.00	0.00

WL: The total wirelength is a weighted sum of the wire length and the number of Vias used, following the same weighting factors in [15], [16].

M1: The M1 metal usage on FFET based cells includes the metal utilization on *both* of FM1 and BM1 layers.

D. Congestion-aware Net Redistribution

This step redistributes nets across the BEOL sides without increasing the number of *DS-nets* which have already been minimized in Step 2. We only consider cells with all input pins connected to 2-pin nets for redistribution since their impact on *DS-nets* is the least. Specifically, on a net-congested local region³, we extract these cells and attempt to change their styles to **PFront** or **PBack** until both sides are equally congestion-balanced. For example, in a local region in Fig. 7(a) where high congestion occurs at the backside, replacing the **PBal** cell in the middle in Fig. 7(a) with **PFront** reduces BS-nets count, resulting in a reduction of congestion on the backside.

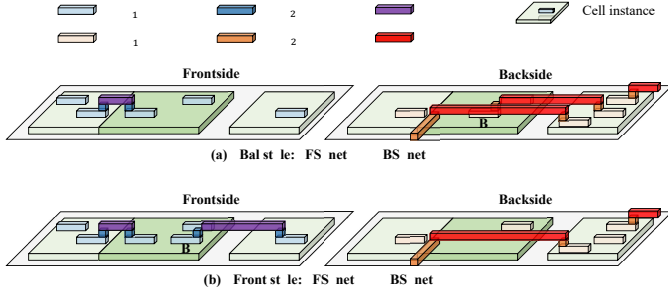


Fig. 7. Example of redistributing nets for mitigating net congestion on both sides.

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

Our FFET based standard cells were implemented in C++ on a Linux system equipped with 128 GB of memory and an AMD EPYC 7452 CPU operating at 2.35 GHz. The input netlists and relevant design rules for synthesizing FFET cells were taken from ASAP 7nm process design kit (PDK) in [18], and our SMT formulation for in-cell routing was solved using Z3 solver package in [19]. We performed logic synthesis using Synopsys Design Compiler [20], and performed block-level placement and routing Cadence Innovus [21].

³We used *net density* as our pre-route congestion metric.

B. Cell Quality Comparison

Table I shows a comparison of the cell metrics between CFET based 4T (4 tracks for routing) standard cells produced by CFET-fp [16] which is available in [22] and follows the same design rules in ASAP 7nm PDK, and FFET based 3.5T cells produced by our proposed cell synthesizer.⁴ In terms of cell area, our FFET cells are 20% smaller on average than that of CFET cells, which is mainly caused by the shorter height of FFET cells (i.e., 3.5T) than CFET cells (i.e., 4T). In addition, FFET cells use 13% less total wirelength than that of CFET cells. In particular, the comparison shows that the in-cell routing for FFET cells is fully completed with the use of FM1 and BM1 only while CFET cells use all metals in M1, M2, and M3. This clearly implies that FFET cells provide a considerable advantage of pin accessibility. (In fact, our cell synthesizer has successfully generated FFET layout cells for 174 netlists in ASAP 7nm PDK, using the metals on the FM1 and BM1 layers only.)

On the other hand, Table II shows a comparison of the cell metrics between CFET based 3T (3 tracks for routing) standard cells produced by CFET-fp [16] following the same design rules in ASAP 7nm PDK, and FFET based 2.5T cells produced by our proposed cell synthesizer.⁵ The comparison shows that FFET cells have a 25% smaller area on average than that of CFET cells while using 31% less total wirelength.

C. Chip Quality Comparison

Table III shows a quality comparison between the chip implementations produced by (*Flow 1*) the conventional non-DTCO physical design flow using the CFET based cells in Table I, (*Flow 2*) the conventional non-DTCO flow using our FFET based cells in Table I, and (*Flow 3*) our proposed DTCO flow using our FFET based cells in Table I for the OpenCore benchmark designs in [23].

⁴For design rules, we set to MAR/EOL/VR/PRL/SHR/MPL = 1/2/1/1/2/1 on 4T CFET cells as well as 3.5T FFET cells.

⁵For design rules, we set MAR/EOL/VR/PRL/SHR/MPL = 1/1/0/1/0/1 on 3T CFET cells as well as 2.5T FFET cells.

TABLE II

COMPARISON OF CELL METRICS BETWEEN CFET BASED 3T STANDARD CELLS PRODUCED BY CFET CELL SYNTHESIZER [16] AND OUR FFET BASED 2.5T STANDARD CELLS. #CPP, WL, M1, M2, M3 INDICATE THE NUMBER OF CONTACTED POLY PITCHES (I.E., CELL WIDTH), TOTAL WIRELENGTH, THE METAL USAGE ON M1, M2, M3, RESPECTIVELY.

Cell	#Cell	#track	CFET based 3T Standard Cells [16]							FFET based 2.5T Standard Cells (Ours)						
			Height	#CPP	Area	WL	M1	M2	M3	Height	#CPP	Area	WL	M1	M2	M3
AND	3	3	144	6.33	912.00	71.00	16.00	11.00	0.00	105	6.33	665.00	54.00	19.33	0.00	0.00
AOI	1	3	144	8.00	1152.00	188.00	35.00	20.00	21.00	105	9.00	945.00	104.00	40.00	0.00	0.00
NAND	4	3	144	11.75	1692.00	236.00	58.25	19.75	19.00	105	12.25	1286.25	146.00	59.00	0.00	0.00
NOR [†]	4	3	144	9.00	1296.00	145.00	37.33	12.67	7.00	105	9.00	945.00	102.00	39.33	0.00	0.00
OR	3	3	144	6.33	912.00	71.00	16.00	11.00	0.00	105	6.33	665.00	54.00	19.33	0.00	0.00
BUF	4	3	144	7.50	1080.00	71.75	19.75	6.00	0.00	105	7.50	787.50	69.25	25.25	0.00	0.00
INV	4	3	144	5.75	828.00	50.50	13.50	6.00	0.00	105	5.75	603.75	48.75	17.75	0.00	0.00
Avg.		3	144	7.81	1124.57	119.04	27.98	12.35	6.71	105	8.02	842.50	82.57	31.43	0.00	0.00
Norm.		3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.73	1.03	0.75	0.69	1.12	0.00	0.00

[†] CFET-fp [16] could not produce CFET cell layout for gate NOR3x2.

TABLE III

CHIP QUALITY COMPARISON BETWEEN THE IMPLEMENTATIONS PRODUCED BY (*Flow 1*) NON-DTCO CONVENTIONAL FLOW USING CFET BASED CELLS, (*Flow 2*) NON-DTCO CONVENTIONAL FLOW USING OUR FFET CELLS, AND (*Flow 3*) OUR DTCO FLOW USING OUR FFET CELLS. *Util.*: CHIP UTILIZATION; *#DRV*: THE NUMBER OF DESIGN RULE VIOLATIONS; *WL*: TOTAL WIRELENGTH.

Design	Util.	#Cell	#Net	(1) Non-DTCO + CFET cells [16]			(2) Non-DTCO + FFET cells (Ours)			(3) DTCO (Ours) + FFET cells (Ours)		
				#DRV	Area	WL	#DRV	Area	WL	#DRV	Area	WL
AES_CIPHER	0.7	10021	10330	8546	10931.35	165475.73	0	8857.82	150115.73	0	8857.82	130193.74
USB_PHY	0.9	387	417	0	512.59	1810.68	0	402.14	2010.14	0	402.14	1914.94
DES3	0.75	51650	51985	1	65416.22	501568.14	3	51334.56	546286.69	4	51334.56	515168.51
FPU	0.9	33369	33666	385	36744.09	278735.02	22	29721.28	324966.30	11	29721.28	292088.43
Avg.				2233	28401.06	236897.39	6.25	22578.95	255844.71	3.75	22578.95	234841.41
Norm.				595.47	1.26	1.01	1.67	1.00	1.09	1.00	1.00	1.00

In summary, *Flow 2* is able to reduce the chip size by 26% over that of *Flow 1* while significantly reducing the DRV count (i.e., 356x). (Note that such a large number of DRVs in *Flow 1*, particularly for AES_CIPHER, is due to the fact that M2 and M3 metal resources have been used up for CFET in-cell routing, as statistically shown in Tables I and II, thereby become very limited M2 and M3 metals available for chip-level routing.

On the other hand, *Flow 3* outperforms *Flow 2* and *Flow 1*. *Flow 3* uses 9% less total wirelength and produces 67% fewer DRVs over that of *Flow 2*. This means that our DTCO flow is very effective in improving chip quality. To validate the effectiveness, we repeatedly applied *Flow 2* and *Flow 3* to design FPU by varying the value of the chip utilization parameter, and compared the implementation results, as shown in Fig. 8. The comparison shows that *Flow 3* consistently reduces the number of DS-nets (dual-sided nets) (by 63% on average), the DRV count (by 51% on average), and total wirelength (by 10% on average) over that of *Flow 2*. Note that our DTCO in *Flow 3* never increases the number of tap-cells over that of *Flow 2* during the (wire-hungry) DS-net minimization in Steps 2 and 3 while *Flow 2* allocates 2, 12, 113, and 0 tap-cells for the implementations of AES_CIPHER, USB_PHY, DES3, and FPU, respectively.

V. CONCLUSION

This work addressed two tasks: (1) FFET (Flip-FET) based standard cell synthesis and (2) DTCO methodology utilizing FFET based cells. A strong and distinct advantage of FFET cells, as opposed to CFET cells, was the convenience of using backside metals as in-cell routes (i.e., no tap-cell overhead) as well as using them as chip routes (i.e., well controllable for tap-cells), the objective of this work was to develop

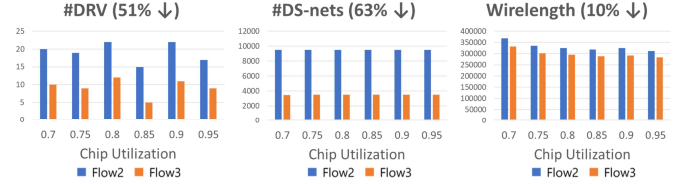


Fig. 8. Design quality comparison of the implementations produced by *Flow 2* (non-DTCO + FFET cells) and *Flow 3* (DTCO + FFET cells) for design FPU by varying the chip utilization value.

an effective solution to the problem of how the backside metals could be fully utilized as interconnects with minimal tap-cell overhead. To this end, we proposed a three-step solution: (1) synthesizing multiple FFET based cells with diverse styles of pin distribution on front/backside BEOLs for each primitive logic gate; (2) performing a tap-cell avoidance cell replacement during placement optimization by exploiting the cells obtained in Step 1; (3) rebalancing the wire usage between the frontside and backside to mitigate net congestion. Experimental results with benchmark circuits showed that our FFET cell-based DTCO flow was able to significantly improve chip quality in terms of chip area, wirelength, and routing failures over that of the CFET/FFET cell-based conventional non-DTCO flows.

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